

Técnicas e Algoritmos em Ciência de Dados

A2 Exam

Answer all the questions.

Time limit: 1.5 hours.

Late submissions will NOT be accepted.

Instructions

Identifier

Please choose a random number of 6 digits and write it at the beginning of your submission. Make sure that you keep a copy of that number (please avoid trivial numbers, such as 000000 or 123456. Also, please avoid numbers starting with zero).

Submission

Submit only one file through ECLASS.

The name of the file will be the random number that identifies you (for example 568423.pdf)

If you have issues, at the very last minute, email your A2 exam as an attachment to alberto.paccanaro@fgv.br with the subject "URGENT – EXAM A2 SUBMISSION". In the body of the message, explain the reason for not submitting through ECLASS.

All the work you submit should be solely your own work. Your submissions will be checked for this.

IMPORTANT: throughout the exam, keep your explanations short and to the point.

Question 1 -- value: 40%

Figure 1 shows a 2-dimensional dataset, with points belonging to 2 different classes (C_0 , C_1) represented by different colours (points from C_0 are red; points from C_1 are blue).

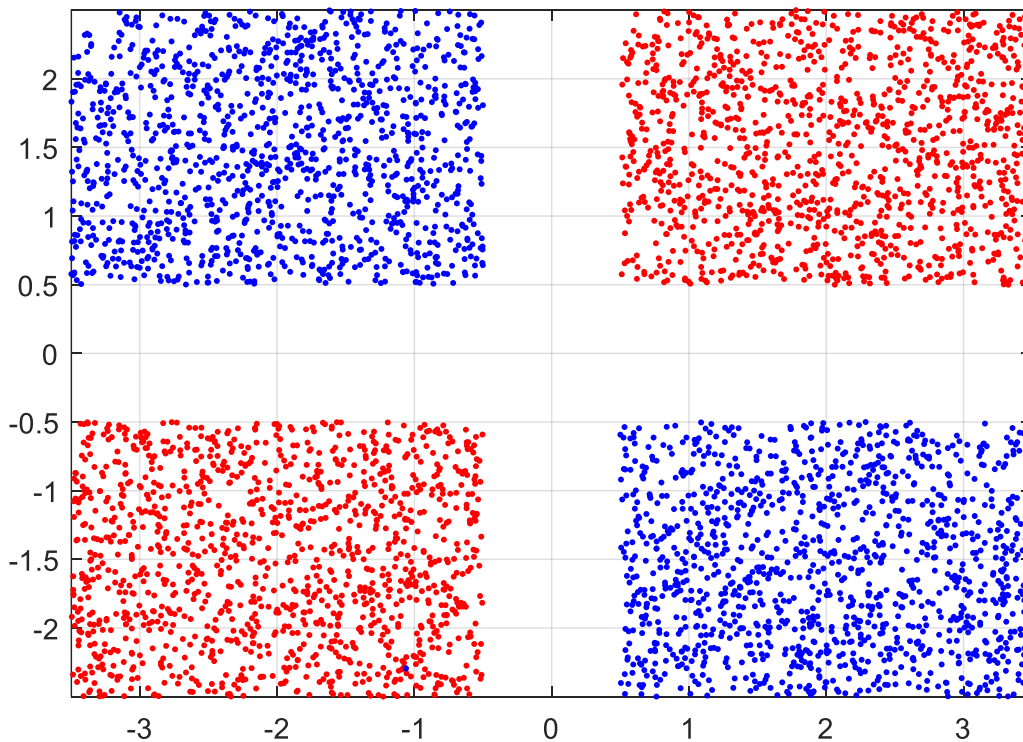


Figure 1

Consider the following classifiers:

- a) Logistic regression
- b) Fisher Linear discriminant, followed by a simple model that discriminates the classes using density estimations by maximum likelihood
- c) Boosted logistic regression
- d) Neural network with sigmoid activation function in the output unit and linear activation function in the hidden units
- e) Neural network with sigmoid activation function in the output unit and also sigmoid activation function in the hidden units
- f) Decision tree
- g) Random Forest

Using *accuracy* as the measure of performance, for each of the classifiers, discuss why you think it should work well, or not, for the task of classifying the datapoints.

Question 2 – value: 20%

Figure 2 shows a 2-dimensional dataset, with points belonging to 3 different classes (C0, C1, C2) represented by different colours (points from C0 are red; points from C1 are blue; points from C2 are green). Note that:

- Each of the three classes has the same number of points.
- Each blob of points is Gaussian distributed.

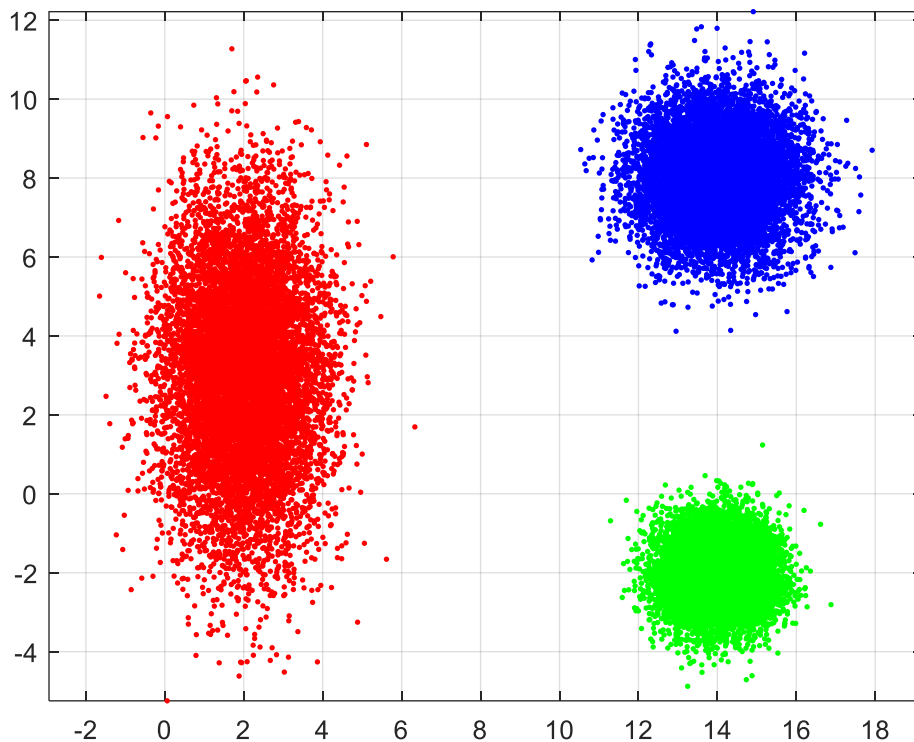


Figure 2

Let us assume that you wish to classify these points using a Decision Tree. Using simply geometrical considerations (no calculations is necessary!):

- draw the decision tree that will be produced by recursive binary splitting when using the Gini index to evaluate the quality of a particular split. On the tree, indicate the splitting condition of each internal node. (Here, only grow the tree, no need for pruning). Provide a detailed justification of your answer.
- draw the decision boundaries that the tree you described in A) would generate. Provide a detailed justification of your answer.

Question 3 – value: 40%

In this exercise you will analyse the derivation of the linear regression model and of the two-class logistic regression models.

- 1) write the equation of the linear regression model
- 2) write the equation of the two-class logistic regression model
- 3) write the sum-of-squares error function. Then show how this error function can be derived for the regression problem (that is, how it is motivated by maximum likelihood).
- 4) write the cross-entropy error function. Then show how this error function can be derived for the two-class classification problem (that is, how it is motivated by maximum likelihood).
- 5) Prove that: the derivative with respect to the weights of the sum-of-squares error function for the linear regression model has a very similar form to the derivative with respect to the weights of the cross-entropy error function for logistic regression.
Note that even if most of the steps of the derivation can be found in the Chris Bishop's book and in the slides, you will have to explain what you do at each step of the derivation and why. (This part is worth 60% of this exercise)

Hint: you will need to use the fact that: $\frac{d\sigma(x)}{dx} = \sigma(x)(1 - \sigma(x))$, where $\sigma(x)$ indicates the sigmoid function.

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